WEEK 6 IMPORTANT QUESTIONS

1. **Explain the need for data splitting in machine learning. Describe the training set, validation set, and test set with a suitable classroom-teaching example.**

2. **Explain overfitting and underfitting in machine learning. Describe their causes, characteristics, and methods to overcome them.**

3. **Explain the concept of regression in machine learning. Describe dependent and independent variables and explain the different types of regression with suitable examples.**

4. **Differentiate between overfitting and underfitting. Analyze their effects on model performance, bias, variance, and generalization. Suggest suitable methods to overcome each problem.**

5. **Compare linear regression, logistic regression, Ridge Regression, Lasso Regression, and Polynomial Regression. Analyze the type of problem each method is suitable for and justify your choices with examples.**

6. **Compare the holdout method, K-fold cross-validation, and Stratified K-fold cross-validation. Analyze their procedures, advantages, limitations, and suitable applications.**

7. **Explain gradient descent in linear regression. Describe the steps involved in finding the best-fit line by minimizing the sum of squared errors.**

8. **Explain the different metrics used to evaluate a regression model. Describe R-squared/Adjusted R-squared, Mean Squared Error (MSE)/Root Mean Squared Error (RMSE).**

9. **Write a Python program to implement Gradient Descent for fitting a straight line to the following data.**

   x = [1,2,3,4,5]

   y = [5,7,9,11,13]

   a. **Initialize slope, intercept, and learning rate.**
   b. **Calculate predicted values and gradients.**
   c. **Update the slope and intercept iteratively.**
   d. **Plot the data points and fitted line.**

10. **Perform data exploration, preprocessing and splitting on datasets Boston Housing price from Scikit learn ?**

11. **Perform data exploration, preprocessing and splitting on datasets Crop yield-past data ?**

12. **Given the data.csv dataset containing temperature and pressure values, compare Linear Regression with Polynomial Regression.**

    a. **Train a Linear Regression model.**
    b. **Train a Polynomial Regression model of degree 4.**
    c. **Plot both models against the actual observations.**

d. **Analyze which model better represents the relationship between temperature and pressure based on the plots.**

13. Build and evaluate a model for predicting sales using the Advertising.csv dataset.

Preprocess the dataset and split it into training and testing sets.

Train a multiple Linear Regression model.

Calculate MSE, RMSE, and R^2 on test data.

Plot actual versus predicted sales and interpret the results.